



ALL



1. (Math Question) Complexity

$f(n) = O(n)$ and $g(n) = O(n!)$, what is the upper bound for $(f+g)(n)$?

Pick **ONE** option

1

☐ $O(n)$

2

☐ $O(n!)$

3

☐ $O(n^3)$

4

☐ $O(n^2)$

5

[Clear Selection](#)

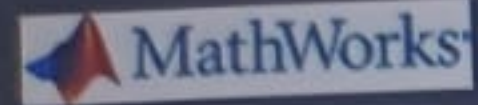
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ALL



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2. (Math Question) Minimum number of cards needs to be drawn

How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen?

Pick **ONE** option

☐ 12

☐ 8

☐ 9

☐ 21

Clear Selection



ENG
IN



3. (Math Question) Committee of 5 members

A committee of 5 people is to be formed from a group of 6 men and 4 women. In how many ways can the committee be formed if it must have at least 2 women?

Pick **ONE** option

☐ 145

☐ 215

☐ 186

☐ 340

Clear Selection



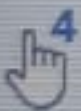
ALL



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4. (Math Question) Find $\cos(x)$ given x

Which of the Following series can be used to calculate $\cos(x)$ for a given x ?

(A) $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{2n!}$

(B) $\sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n!}$

(C) $\sum_{n=0}^{\infty} \frac{x^{2n}}{2n!}$

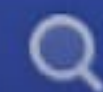
(D) $\sum_{n=0}^{\infty} \frac{(-x)^n}{n!}$

Pick **ONE** option

<https://www.hackerrank.com/test/540omemi024/questions/btn6pfhp9qd>



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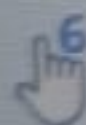


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5. (Math Question) Largest divisor

What is the largest integer n for which 4^n divides $1000!$?

Pick **ONE** option

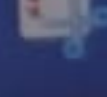
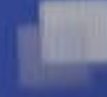
☐ 124

☐ 249

☐ None of these

☐ 995

Clear Selection





ALL



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6. (Math Question) Integral Multiple

For what value of p is $a^p - a$ an integral multiple of p if a is an integer?

Pick **ONE** option

☐ 27

☐ 17

☐ Both 27 and 17

☐ Neither 27 nor 17

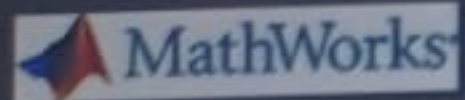
Clear Selection



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ALL



4

Pick **ONE** option

5

☐ 83

6

☐ 125

7

☐ 124

8

☐ 84

9

Clear Selection

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8. (Prog. Concept Question) Quicksort

Let P be a QuickSort Program to sort numbers in ascending order using the first element as pivot. Let t_1 and t_2 be the number of comparisons made by P for the inputs $\{1, 2, 3, 4, 5\}$ and $\{4, 1, 5, 3, 2\}$ respectively. Which one of the following holds?

Pick **ONE** option

☐ $t_1=5$

☐ $t_1=t_2$

☐ $t_1>t_2$

☐ $t_1<t_2$

Clear Selection



ALL



6

Pick **ONE** option

7

☐ 1

8

☐ 2

9

☐ 3



11

Clear Selection

9. (Prog. Concept Question) Processes

Consider three CPU-intensive processes, which require 30, 20 and 10 time units and arrive at times 0, 5 and 10, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at the end.

10. (Prog. Concept Question) Element Location in Column Major Order

A 2-D array, $A[\text{row}][\text{col}]$ is stored in column major order. Base address of A is denoted by $\text{BASE}(A)$ and W is the number of bytes required to store a single element of array A . Then the memory location of $A[J][K]$ is computed as follows,

Pick **ONE** option

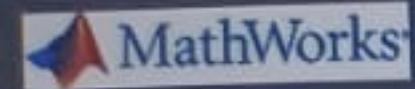
☐ $\text{LOC}(A[J, K]) = \text{Base}(A) + W [\text{row} \times (K-1) + (J-1)]$

☐ $\text{LOC}(A[J, K]) = \text{Base}(A) + W [\text{col} \times (K-1) + (J-1)]$

☐ $\text{LOC}(A[J, K]) = \text{Base}(A) + W [\text{col} \times (J-1) + (K-1)]$

☐ $\text{LOC}(A[J, K]) = \text{Base}(A) + W [\text{row} \times (J-1) + (K-1)]$

Clear Selection



ALL



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11. (Programming Fundamentals) Find the Time Complexity

What is the least time complexity required to find the missing minimum positive integer in an array of size n with constant space?

Pick **ONE** option

☐ $O(n \cdot \log n)$

☐ $O(n \cdot \sqrt{n})$

☐ $O(n)$

☐ $O(1)$

Clear Selection



12. (Prog. Concept Question) SQL Query Identification

Consider the following relational schema

Suppliers(sid:integer, sname:string, city:string, street:string)
Parts(pid:integer, pname:string, color:string)
Catalog(sid:integer, pid:integer, cost:real)

Consider the following relational query on the above database and identify:

```
SELECT S.sname FROM Suppliers S WHERE S.sid NOT IN (SELECT C.sid FROM Catalog C WHERE C.pid NOT IN (SELECT P.pid FROM Parts P WHERE P.color <> 'blue'))
```

Pick **ONE** option

- ☐ Find the names of all suppliers who have supplied a non-blue part.
- ☐ Find the names of all suppliers who have not supplied a non-blue part.
- ☐ Find the names of all suppliers who have supplied only blue parts.



ALL



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13. (C Question) Character Pointer

Predict the output of the given program.

```
#include <stdio.h>

int main(void)
{
    char* s = "B\x4b";
    printf("%s", s);
    return 0;
}
```

Pick **ONE** option

<https://www.hackerrank.com/test/540omemi024/questions/4rca9mi5sj0>





ALL



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14. (C Question) C Preprocessor

What is the output of the following C program?

```
#define ADD(x) x + x
```

```
#define SUB(x) x - x
```

```
int main()
```

```
{
```

```
    int y = ADD(3) / SUB(3);
```

```
    printf("%d", ADD(y));
```

```
}
```

Pick **ONE** option

☐ 2☐ Divide by zero exception at runtime☐ 1

<https://www.hackerrank.com/test/540omemi024/questions/4pqfi07trr9>

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ALL



15. (C Question) Pointers and References in C

Will the below code print the incremented value of the local variable a?

12

```
#include <stdio.h>
```

```
int* inc(int val)
```

```
{
```

```
int a = val;
```

13

```
a++;
```

```
return &a;
```

```
}
```

14

```
int main(void)
```

```
{
```

```
int a = 10;
```

15

```
int *val = inc(a);
```

```
printf("\n Incremented value is equal to [%d] \n", *val);
```

```
return 0;
```

16

```
}
```

17

Pick **ONE** option



YES

<https://www.hackerrank.com/test/540omemi024/questions/eifpm1na90f>



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16. (C Question) sizeof Question

What will be the output of the following C code snippet?

```
int main(){  
    printf("%d\n",sizeof('P'));  
}
```

Pick **ONE** option

☐ 1☐ 4☐ 2

<https://www.hackerrank.com/test/540omemi024/questions/9b0t1fapjfs>



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17. (C Question) C Output question

ALL

What will be printed in the console?



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```
#include <stdio.h>
```

```
int f1() { printf ("MATLAB"); return 1;}
```

```
int f2() { printf ("Simulink"); return 1;}
```

```
int main()
```

```
{
```

```
    int p = f1() + f2();
```

```
    return 0;
```

```
}
```

Pick ONE option





ALL



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18. (C Question) Dynamic memory allocation

With respect to the differences between malloc and calloc, select the option(s) that are true.

Pick **ONE OR MORE** options

- ☐ Calloc initializes the allocated memory with 0 (zero) while malloc just fills it with garbage
- ☐ Calloc always allocates contiguous memory locations but malloc may or may not allocate contiguous memory locations
- ☐ Calloc takes the number and type of data as arguments while malloc takes the number of bytes
- ☐ Calloc may or may not allocate contiguous memory locations but malloc always allocates contiguous memory locations

Clear Selection

<https://www.hackerrank.com/test/540omemi024/questions/9q820od5rn9>

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19. (C Question) Pointers II

Predict the output of following C program:

```
# include <stdio.h>

void main ( )

{

    char *s[ ] = {"black", "white", "yellow", "violet"};

    char **ptr[] = {s+3, s+2, s+1, s}, ***p;

    p = ptr;

    ***p;

    printf("%s\n",*--***p + 3);

}
```

Pick **ONE** option

Predict the output of following program:

```
#include<stdio.h>

void main()

{

int a[]={1,2,4,6,8};

int* p[]={a,a+1,a+2,a+3,a+4};

int** p1=p;

int* p2=*(p+2);

printf("%u %u %u\n",+++p2,++*p2++,+++++p1);

}
```

Pick **ONE** option

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ALL



24. (C++ Question) "new" vs "malloc" in CPP

Select all statements from the following which are TRUE:

Pick **ONE OR MORE** options

21

☐ "new" calls constructor while "malloc" does not.

22

☐ "new" is an operator, while "malloc" is a function.

23

☐ "new" can be overloaded, while "malloc" cannot be.

24

☐ "new" returns exact data type, while "malloc" returns a void pointer.

25

☐ "new" returns NULL if memory is unavailable, while "malloc" throws an exception.

26

Clear Selection

27





ALL



25. (C++ Question) Increment Operator

Choose the correct option regarding the C++ increment operator:

Pick **ONE** option

☐ ++i executes faster than i++

☐ i++ executes faster than ++i

☐ Time taken by ++i and i++ to execute is same

☐ Depends on the compiler

Clear Selection



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ALL



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26. (C++ Question) Conversion Constructor

Which of the following are true about conversion constructor in c++?

Pick **ONE OR MORE** options

☐ Conversion constructor can be used to convert data types of public variables in a class

☐ Conversion constructor accepts one argument of a different type

☐ Conversion constructor does not exist in c++

☐ Conversion constructor can do the job of the destructor in C++

Clear Selection



28. (C++ Question) unsigned char Output

ALL



What would the following code output?

```
#include<iostream>

int main()
{
    char c;

    c = 128;

    printf("%d %d\n", c, (unsigned char)c);

    return 0;
}
```

Pick **ONE** option





ALL



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29. JavaScript: Appending Element to the Array

```
let arr = [1, 2, 3];  
// complete the missing Javascript code on this line  
console.log(arr); // outputs [1, 2, 3, 4]
```

Pick **ONE** option

☐ arr[arr.length + 1] = value

☐ arr[arr.length] = value

☐ arr[arr.length - 1] = value

☐ arr = arr + value

Clear Selection





ALL



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30. Which of the following is considered an object in Javascript?

Which of the following is considered an object in Javascript?

Pick **ONE OR MORE** options

☐

function

☐

true

☐

10

☐

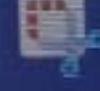
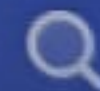
string

Clear Selection



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31. JSON data format

JSON uses a few strict rules to encode its content. Which of the following are true about those rules? Choose one or more.

Pick **ONE OR MORE** options

☐ JSON uses double quotation marks to delimit strings.

☐ JSON uses single quotation marks to delimit strings.

☐ Trailing commas are forbidden.

☐ Trailing commas are allowed and recommended because of cleaner diffs when working with version control systems.

Clear Selection

ALL

32. Function Borrowing

What is the output of the following code?

```
const person1 = {  
  name: 'John Doe',  
  age: 30,  
  getDetails() {  
    return `${this.name} is ${this.age} years old.`;  
  }  
};  
  
const person2 = {  
  name: 'Jane Doe',  
  age: 25  
};  
  
console.log(person1.getDetails.call(person2));  
console.log(person1.getDetails.apply(person2));
```


33. Clean Conditional

What is a good way to clean the conditional expression?

```
...  
30 if(user.experience > 6 && user.earnings >= 5000 && user.contracts > 60 &&  
    user.warnings == 0){  
31   sendInvitation(user)  
    user.risingTalent = true  
    }  
32 ...
```

Pick **ONE** option

☐ Each component of the expression should be moved to a separate if statement.

☐ Remove 'user.warnings == 0' from the expression.



ALL



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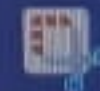
37

```
function test (a, b=1, c, ...args)
{ }
```

```
console.log(test.length)
```

Pick **ONE** option☐ 4☐ 3☐ 2☐ 1

Clear Selection





ALL



35. (JavaScript) this

What will be the output?

```
const object = {  
  abc: function () {  
    return this;  
  },  
  
  def: function () {  
    return function () {  
      return this;  
    }  
  }  
};  
  
console.log(object.abc());  
console.log(object.def()());
```

Pick **ONE** option

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36. Invoked Function Expression

ALL



What is the output of the following code?

```
33  const func = (function () {  
34      let counter = 0;  
35      return function () {  
36          return counter++;  
37      }  
38  })();
```

```
39  let result = func();  
40  result = func();  
41  result = func();  
42  console.log(result);
```

37

38

Pick **ONE** option

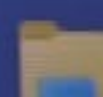
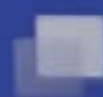
39



Error



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ALL



39. (Python Question) Is alpha

What is the output of the following?

```
print('a B'.isalpha())
```

Pick **ONE** option

☐ True

☐ False

☐ None

40. (Python Question) One for all, all for one

What is the expected out of the function shown below?

```
all([2.56,4,0,5.12])
```

Pick **ONE** option

☐ False

☐ error

☐ True

☐ 0

41. (Python Question) Decorate the function

What is the output of the code shown below?

```
def f(x):  
    def f1(a, b):  
        print("hello")  
        if b==0:  
            print("NO")  
        return  
    return f(a, b)  
    return f1  
@f  
def f(a, b):  
    return a%b  
f(4,0)
```


42. (Python Question) Set

What is the output of the following code?

```
>>> a={4,5,6}  
>>> b={2,8,6}  
>>> a+b
```

Pick **ONE** option

☐ {4,5,6,2,8}

☐ {2, 4, 5, 6, 8}

☐ {6,13,12}

3

43. (Python Question) Game of append

What is the expected output?

```
def foo(i, x=[]):  
    x.append(x.append(i))  
    return x  
for i in range(3):  
    y = foo(i)  
    print(y)
```

Pick **ONE** option

☐ [[[0]], [[[0]], [1]], [[[0]], [[[0]], [1]], [2]]]

44. (Python Question) The power of map

What is the output of the following?

```
elements = [0, 1, 2]
def incr(x):
    return x+1
print(list(map(incr, elements)))
```

Pick **ONE** option

☐ [1, 2, 3]